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SBA-Agentic — Single Control Plane for AI-Driven Business Operations

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1. Pendahuluan dan Latar Belakang

SBA-Agentic menghadirkan Single Control Plane untuk operasi bisnis berbasis AI. Dokumen ini menyatukan arsitektur, spesifikasi use-case, implementasi teknis di `apps/app`, panduan integrasi, serta deliverables operasional.

Dokumentasi ini disusun untuk menjadi pedoman yang dapat langsung dipakai lintas peran (product, engineering, ops) agar investasi pengembangan tetap terarah, dapat diaudit, dan

dapat dioperasikan.

Terminologi konsisten mengikuti pedoman di `docs/README.md`, termasuk keamanan (CSP, RBAC), observability (Prometheus/OTel), dan readiness produk.

1.1 Tujuan Dokumen

- Menetapkan definisi dan ruang lingkup kontrol plane SBA (produk, teknis, operasional).
- Menyediakan artefak arsitektur yang terverifikasi terhadap implementasi `apps/app`.
- Menjadi referensi tunggal untuk integrasi (env, endpoint, otorisasi) dan standar operasional (metrics/health).

1.2 Ruang Lingkup

In-scope (berdasarkan `docs/Use-Case Specification apps-app.md`):

- Tenant & Workspace experience
- Agent lifecycle, reasoning UI, interrupt & control
- Run/workflow control dan observability
- Analytics & monitoring UX
- Onboarding & adoption, aksesibilitas, performa

Out-of-scope:

- Model training internal
- Infrastruktur runtime agent di luar UI/API edge

1.3 Definisi dan Terminologi

- Control Plane: lapisan koordinasi UX, governance, observability, dan kontrol agentic untuk operasi bisnis.
- Tenant: boundary multi-tenant untuk konfigurasi, data, dan kebijakan.
- Workspace: konteks kerja di dalam tenant untuk tim/proyek.
- Agent: unit AI yang dapat diajak berinteraksi dan menjalankan tugas.
- Run: eksekusi workflow/agent task yang dapat diobservasi (logs, metrics, status).
- AG-UI: protokol antarmuka agentic untuk streaming message/reasoning/interrupt/meta-event.

2. Arsitektur Sistem dan Kontrol Plane

2.1 Ikhtisar

Kontrol plane mengoordinasikan UX, Reasoning, Observability, dan Integrasi Backend. `apps/app` bertindak sebagai edge untuk API Next.js dan permukaan produk utama.

2.2 Komponen Utama

- UX Shell (layout, routing, auth, i18n)
- Product Features (dashboard, agents, runs, analytics)

- Real-time Interfaces (SSE/WS/AG-UI)
- Observability (metrics, tracing)
- API Edge (Next.js route handlers)

2.3 Integrasi Backend Produksi

- Supabase Auth dan Data: env publik (`NEXT\_PUBLIC\_SUPABASE\_\*`) dan klien server (`@sba/supabase/clients/server`)
- Observability: Prometheus/OTel, endpoints `/api/metrics/json|prometheus`
- Keamanan: CSP nonce, RBAC guard, rate limiting

2.4 Diagram Arsitektur

```

flowchart TD
    User((User)) -->|Auth| App[apps/app]

    subgraph ControlPlane
        App --> UX[UX Shell]
        App --> AG[Agents & Runs]
        App --> AN[Analytics]
        App --> OBS[Observability]
        App --> API[Next.js API Edge]
    end

    end

    API --> SUP[(Supabase Auth & Data)]

    OBS --> PROM[(Prometheus/OTel)]

    AG --> ORCH[Orchestrator]

    ORCH --> RUBE[@sba/rube]

```

2.4.1 Diagram Aliran Data Bisnis (Tenant → Workspace → Run)

```

flowchart LR
    U[User] -->|login| AUTH[Auth Session + JWT]
    AUTH -->|select| T[Tenant]
    T -->|context| W[Workspace]
    W -->|create/execute| R[Run]

```

```

subgraph CP[apps/app Control Plane]

  AUTH --> UI[UX Shell]

  UI --> AGUI[AG-UI Streaming]

  UI --> GOV[RBAC Guard]

  UI --> API[API Edge]

  API --> OBS[Metrics/Tracing]

end

API -->|read/write| DB[(Supabase Data)]

API -->|cache/limit| KV[(Upstash Redis)]

R -->|events/logs| AGUI

OBS -->|scrape| PROM[(Prometheus/OTel)]

```

3. Spesifikasi Use-Case (apps/app)

Ringkasan dan spesifikasi diadaptasi dari `docs/Use-Case Specification apps-app.md`.

3.1 Ringkasan Eksekutif

“Single Control Plane for AI-Driven Business Operations”

`apps/app` adalah orchestrator pengalaman bisnis berbasis AI: tenant/workspace/agent/run/knowledge, real-time reasoning, analytics, dan observability.

3.2 Context & Scope

```

apps/app

├── UX Shell (layout, routing, auth, i18n)

├── Product Features (dashboard, agents, runs, analytics)

├── Real-time Interfaces (SSE, WS, AG-UI)

├── Product Telemetry (metrics, observability)

└── API Edge (Next.js route handlers)

```

3.3 Personas & Aktor

- Business Owner — automasi & insight
- Ops Manager — monitoring & reliability
- AI Operator — control agent & workflow
- Admin — governance & security

3.4 Use-Case Utama

- UC-01 Authenticate & Enter Workspace
- UC-02 Navigate Product Domains
- UC-03 Manage Agents & Conversations
- UC-04 Execute & Observe Runs
- UC-05 Analyze Business Metrics
- UC-06 Monitor System Health
- UC-07 Onboarding & Activation
- UC-08 Governance, RBAC & Audit

3.5 Ringkasan Use-Case (Detail)

Bagian ini merangkum spesifikasi detail yang bersumber dari `docs/Use-Case Specification apps-app.md`.

UC-01 Authenticate & Enter Workspace

- Alur: login → session/JWT → pilih workspace → set context (store + header) → guard RBAC aktif.
- Target UX: state tanpa ambiguitas, error recovery jelas, form aksesibel.
- Rujukan: Bagian 6.1 dan Bagian 7.5.

UC-02 Navigate Product Domains

- Alur: navigasi antar domain produk tanpa kehilangan konteks tenant/workspace.
- Target UX: keyboard accessible, hirarki prediktabel, tidak ada full reload.

UC-03 Manage Agents & Conversations

- Alur: create agent → mulai percakapan → reasoning streamed (SSE/WS) → UI update real-time.
- Diferensiasi: reasoning transparency, interrupt & control, progressive disclosure.
- Rujukan: Bagian 4.3 dan Bagian 6.2.

UC-04 Execute & Observe Runs

- Alur: eksekusi workflow/run → live logs + status semantics → failure reason surfaced.
- Target UX: tidak ada silent failure, timeline-based logs.

UC-05 Analyze Business Metrics

- Peran: decision support (bukan sekadar grafik).

- Output: metrik usage/performance/KPI bisnis, ekspor report.

UC-06 Monitor System Health

- Alur: lihat health/metrics → alert visibility → no false calm.
- Rujukan: Bagian 4.4, Bagian 5.2, Bagian 8.

UC-07 Onboarding & Activation

- Strategi: langkah progresif, feedback sukses jelas, boleh skip.
- Metrik: activation rate dan time-to-first-value.

UC-08 Governance, RBAC & Audit

- Prinsip: security visible, not obstructive.
- Mekanisme: RBAC guard, audit store, security headers.

4. Implementasi Teknis (apps/app)

4.1 Struktur Kode

Struktur utama `apps/app` menyertakan UX Shell, fitur produk (agents, runs, analytics), integrasi real-time, dan API edge.

Rujukan operasional:

- Kebijakan impor UI dan panduan a11y/E2E: `apps/app/README.md`

Ringkasan modul (evidence: struktur folder `apps/app/src/\*`):

- `src/app/\*`: routing Next.js App Router, halaman domain, dan route handlers API.
- `src/features/\*`: fitur produk (agentic, run-controls, analytics, onboarding).
- `src/domains/\*`: domain logic (auth, workspace).
- `src/entities/\*`: state & model UI untuk tenant/workspace/run/dll.
- `src/shared/\*`: util lintas fitur (rbac guard, metrics registry, konfigurasi).
- `src/lib/\*`: integrasi protokol dan helper (AG-UI, supabase, test utils).

4.2 Alur Kerja Bisnis yang Diotomasi

Kontrol plane memfasilitasi alur kerja utama berikut:

- Orkestrasi percakapan agentic dan kontrol eksekusi (create agent → run → observasi hasil).
- Monitoring health dan metrik aplikasi untuk reliability operasi.
- Query/ingest knowledge untuk mendukung reasoning dan pencarian kontekstual.
- Workflow dan task scheduling untuk automasi proses bisnis.

4.3 Komponen AI dan Reasoning

Komponen AI antarmuka berfokus pada transparansi dan kontrol:

- AG-UI protocol: message, multimodal content, reasoning steps, interrupts, meta-events (`apps/app/src/lib/agui/protocol.ts`).
- Real-time reasoning: streaming via SSE/WS, UI mengikat event stream untuk state mesin percakapan.
- Kontrol eksekusi: interrupt/pause/stop untuk memastikan user dapat memegang kendali.

4.4 Observability & Metrics

Observability menjadi kontrak utama kontrol plane untuk operasi produksi:

- Metrics (RBAC `analytics:read`):
- `GET /api/metrics/json` menyediakan snapshot counters/errors/histograms serta `p95/p99`.
- `GET /api/metrics/prometheus` menyediakan format Prometheus dan gauge ringkas (p95/p99/error rate/throughput).
- Health:
- `GET /api/health` menyediakan status cepat readiness/liveness.
- Tenant labeling:
- Header `x-tenant-id` dinormalisasi oleh util `ensureTenantHeader` agar label tenant konsisten.

4.5 Keamanan & RBAC

Keamanan kontrol plane menggabungkan autentikasi, RBAC, dan security headers:

- Autentikasi: session/JWT via Supabase.
- Otorisasi: guard RBAC pada endpoint sensitif menggunakan `withPermissions` dan/atau `withRBAC`.
- Tenant isolation: konteks tenant wajib tersedia melalui header `x-tenant-id`.
- Rate limiting: Upstash Redis untuk proteksi endpoint publik dan auth.

5. Panduan Integrasi Sistem

5.1 Konfigurasi Lingkungan

Variabel environment minimal mengacu ke `docs/README.md`:

- `NEXT\_PUBLIC\_APP\_URL`
- `NEXT\_PUBLIC\_SUPABASE\_URL`
- `NEXT\_PUBLIC\_SUPABASE\_ANON\_KEY`
- `UPSTASH\_REDIS\_REST\_URL`
- `UPSTASH\_REDIS\_REST\_TOKEN`

Untuk staging, rujuk `docs/DEPLOYMENT\_STAGING.md` dan playbook go-live di `docs/README.md`.

5.2 API & Endpoints

Spesifikasi ringkas endpoint kontrol plane (tidak menggantikan OpenAPI penuh):

Area	Method	Endpoint	Otorisasi	Catatan
Health	GET	`/api/health`	Publik	Readiness/liveness cepat
Metrics	GET	`/api/metrics/json`	RBAC `analytics:read`	Snapshot metrik (JSON)
Metrics	GET	`/api/metrics/prometheus`	RBAC `analytics:read`	Export Prometheus
Docs	GET	`/api/openapi`	Permission `system.docs`	Dokumen OpenAPI (JSON)
Auth	POST	`/api/auth/login`	Publik	Login Supabase (UC-01)
Knowledge	POST	`/api/knowledge/search`	Sesuai kebijakan	Search knowledge + caching
Runs	GET/POST	`/api/runs`	Sesuai kebijakan	List/create runs
Workflows	GET/POST	`/api/workflows`	Sesuai kebijakan	List/create workflows

Catatan tenant:

- Header `x-tenant-id` wajib pada operasi multi-tenant; util `ensureTenantHeader` menormalisasi nilai kosong/hilang menjadi `unknown`.

5.3 Model Data Bisnis

Model data bisnis yang terefleksi di kontrol plane (ringkas):

- Tenant: konfigurasi tenant dan batasan fitur/limit (lihat store tenant).
- Workspace: konteks kerja termasuk membership dan flags fitur (chat/automation/timeline) (lihat store workspace).
- Agent: konfigurasi agent dan state percakapan.
- Run: status eksekusi workflow dan telemetry (logs/metrics/events).
- Knowledge: dokumen/embeddings/index untuk retrieval.

Sumber teknis:

- State/model UI tenant: `apps/app/src/entities/tenant/model.ts`
- State/model UI workspace: `apps/app/src/entities/workspace/model.ts`
- Endpoint dan integrasi: `apps/app/src/app/api/\*/route.ts`

5.4 Mekanisme Otorisasi

Mekanisme otorisasi menggabungkan Supabase session dan RBAC guard:

- Supabase menyediakan session/JWT; role user dapat dipropagasikan dari server session.
- RBAC guard mengevaluasi permission terhadap role untuk endpoint tertentu.
- Mode test/dev dapat memakai cookie `\_\_test\_auth=admin` untuk mem-bypass atau menyuntikkan role sesuai kebijakan lingkungan.

6. Diagram Sequence

6.1 Alur Bisnis Utama — Authenticate & Enter Workspace

```
sequenceDiagram
    participant U as User
    participant A as apps/app
    participant S as Supabase

    U->>A: Open login
    A->>S: Auth (email/password, OAuth)
    S-->>A: Session + JWT
    A->>U: Redirect to (authenticated)
    U->>A: Select Workspace
    A->>A: Set context (store + header)
```

6.2 Pengambilan Keputusan AI — Agent Run

```
sequenceDiagram
    participant U as User
    participant A as apps/app
    participant O as Orchestrator
    participant R as @sba/rube

    U->>A: Start conversation
    A->>O: Submit task
```

0->>R: Enforce policies (RBAC)

0-->>A: Stream reasoning (SSE)

A-->>U: Update UI real-time

6.3 Error Handling & Retry — Exponential Backoff

```
sequenceDiagram
    participant O as Orchestrator
    participant T as ToolRegistry

    O->>T: Execute tool
    T-->>O: Failure (retryable)
    O->>O: Schedule retry (2^n seconds)
    O->>T: Retry until max
```

7. Manajemen Produk & UX

7.1 Roadmap Pengembangan

- Now: Reliability, UX clarity, adoption
- Next: Monetization hooks, pricing UX
- Later: Marketplace, extensibility

7.2 Matriks Prioritas Fitur

Fitur	Nilai Bisnis	Kompleksitas	Prioritas
Agents	Tinggi	Sedang	P1
Runs	Tinggi	Sedang	P1
Analytics	Tinggi	Tinggi	P1
Onboarding	Sedang	Rendah	P2

7.3 Metrik Keberhasilan

- Activation rate, Time-to-First-Value
- Error rate ≤0.5%, p95 latency ≤500ms

7.4 Prinsip UX & Aksesibilitas

- Continuity, Explainability, Control, Observability, Accessibility (WCAG)

7.6 Matriks Keterlacakan (Ringkas)

Matriks ini mengikat kebutuhan bisnis (use-case) ke bukti implementasi. Versi lengkap tersedia pada ``docs/TRACEABILITY_MATRIX.md`` dan ``docs/requirements_traceability.md``.

Use-Case	Bukti Implementasi	Kontrak Operasional	
UC-01 Authenticate & Enter Workspace	<code>`apps/app/src/app/(public)/*`, `apps/app/src/domains/auth/*`</code>	<code>`GET /api/health`</code> untuk readiness, audit & error metrics	
UC-03 Manage Agents & Conversations	<code>`apps/app/src/features/agentik/*`, `apps/app/src/lib/agui/protocol.ts`</code>	Streaming SSE/WS + metrik route per tenant	
UC-04 Execute & Observe Runs	<code>`apps/app/src/features/run-controls/*`, `apps/app/src/app/api/runs/*/route.ts`</code>	Timeline/logs + error surfaced + histogram latency	
UC-05 Analyze Business Metrics	<code>`apps/app/src/features/analytics/*`, `apps/app/src/app/api/metrics/*/route.ts`</code>	<code>`GET /api/metrics/json`</code>	<code>prometheus` (RBAC `analytics:read`)</code>
UC-08 Governance, RBAC & Audit	<code>`apps/app/src/shared/utlis/rbac-guard.ts`, `apps/app/src/shared/metrics-registry.ts`</code>	Tenant header, RBAC guard, security headers, audit store	

7.5 Visualisasi UX — UC-01 Login & Workspace Selector

Skenario ini dipilih karena menjadi pintu masuk kontrol plane, memerlukan navigasi yang jelas, elemen antarmuka utama yang eksplisit (form input, CTA, error), serta state interaksi yang terdefinisi untuk menghindari ambiguitas.

7.5.1 Wireframe (Responsive + States)

Sumber: ``workspace/03_Design-System/wireframes/sba-uc01-login-workspace-wireframe.mmd``

```
flowchart TB
```

```

classDef frame fill:#ffffff,stroke:#111827,stroke-width:2px;

classDef panel fill:#f9fafb,stroke:#374151,stroke-width:1px;

classDef content fill:#ffffff,stroke:#9ca3af,stroke-dasharray: 4 3;

classDef btn fill:#e5e7eb,stroke:#111827,stroke-width:1px;

classDef btnHover fill:#dbeafe,stroke:#1d4ed8,stroke-width:2px;

classDef btnActive fill:#bfdbfe,stroke:#1d4ed8,stroke-width:3px;

classDef badge fill:#111827,stroke:#111827,color:#ffffff;

classDef error fill:#fee2e2,stroke:#b91c1c,stroke-width:2px;

classDef focus fill:#ecfeff,stroke:#0891b2,stroke-width:2px;

```

```
subgraph D[Desktop >= 1024px]
```

```
    direction TB
```

```
    DFrame[App Shell]
```

```
    DFrame --> DH[Header: Logo • Workspace • Profile]
```

```
    DFrame --> DB[Body]
```

```
    DB --> DSide[Sidebar: Dashboard / Agents / Runs / Analytics / Monitoring]
```

```
    DB --> DMain[Main Content Area]
```

```
    DMain --> DTitle[Title: Select Workspace]
```

```
    DMain --> DSearch[Search / Filter]
```

```
    DMain --> DList[Workspace List - cards]
```

```
    DList --> DCard1[Workspace Card: Name + Tenant Badge]
```

```
    DList --> DCard2[Workspace Card: Name + Tenant Badge]
```

```
    DMain --> DCTA[Primary Button: Enter Workspace]
```

```
    DMain --> DHelp[Secondary: Help / Docs]
```

```
end
```

```
subgraph M[Mobile <= 480px]
```

```

direction TB

MFrame[App Shell]

MFrame --> MH[Header: Logo • Menu • Profile]

MFrame --> MMain[Main]

MMain --> MTitle[Title: Select Workspace]

MMain --> MSearch[Search]

MMain --> MList[Workspace List - stacked]

MList --> MCard1[Workspace Row: Name + Tenant Badge]

MList --> MCard2[Workspace Row: Name + Tenant Badge]

MMain --> MCTA[Primary Button: Enter]

end


subgraph S[States - Interactive]

direction TB

S0[Default State]

S1[Hover State]

S2[Active/Pressed State]

S3[Error State]

S4[Focus State - Keyboard]


BtnDefault[CTA Button]

BtnHover[CTA Button]

BtnActive[CTA Button]

ErrBanner[Inline Error Banner: Login failed • Workspace not accessible]

FieldFocus[Focused Input: Workspace Search]


S0 --> BtnDefault

S1 --> BtnHover

S2 --> BtnActive

```

```

S3 --> ErrBanner

S4 --> FieldFocus

end

class DFrame,MFrame frame;

class DH,DB,DMain,MH,MMain panel;

class DSide,DTitle,DSearch,DCard1,DCard2,DHelp,MTitle,MSearch,MCard1,MCard2 content;

class DList,MList panel;

class DCTA,MCTA,BtnDefault btn;

class BtnHover btnHover;

class BtnActive btnActive;

class S0,S1,S2,S3,S4 badge;

class ErrBanner error;

class FieldFocus focus;

```

7.5.2 Sequence Diagram — Error Recovery & RBAC

Sumber: `workspace/02\_Architecture/diagrams/sba-sequence-uc01-error-recovery.mmd`

```

sequenceDiagram
    autonumber
    participant U as User
    participant UI as apps/app (UI)
    participant API as apps/app (API Edge)
    participant S as Supabase Auth
    participant RBAC as RBAC Guard
    participant W as Workspace Service
    participant OBS as Metrics/Logging

    U->>UI: Open /login
    UI->>U: Render login form

```



```
U->>UI: Submit credentials

UI->>API: POST /api/auth/login

API->>S: Authenticate

alt Auth success

    S-->>API: Session + JWT

    API->>OBS: Log login_success

    API-->>UI: 200 OK (session)

    UI->>RBAC: Evaluate role/permissions

    RBAC-->>UI: Allow

    UI->>API: GET /api/workspaces (tenant context)

    API->>W: List workspaces for user

    W-->>API: Workspaces

    API->>OBS: Record latency + count

    API-->>UI: 200 OK (workspaces)

    UI->>U: Render workspace selector

else Invalid credentials

    S-->>API: 401 Unauthorized

    API->>OBS: Log auth_failure + count

    API-->>UI: 401 (error)

    UI->>U: Show inline error + recovery CTA

else Network / rate limit

    API-->>UI: 429/503 (retryable)

    UI->>OBS: Log retry_scheduled

    UI->>UI: Exponential backoff retry (bounded)

    UI->>U: Show "Retrying..." state

end

opt Workspace not accessible (RBAC)

    RBAC-->>UI: Deny (insufficient permission)
```

```

UI->>OBS: Log rbac_denied

UI->>U: Show access denied + contact admin

end

```

7.5.3 Validasi UX

Status: pending sign-off UX.

Checklist UAT (UX):

- Layout desktop ($\geq 1024\text{px}$) menampilkan header, sidebar, list workspace, dan CTA utama.
- Layout mobile ($\leq 480\text{px}$) menampilkan header ringkas, list workspace stacked, dan CTA utama.
- States minimum terdefinisi: default, hover, active/pressed, error, focus (keyboard).
- Copy error tidak ambigu dan menyediakan recovery action.
- Aksesibilitas: fokus keyboard terlihat, urutan tab masuk akal, kontras CTA memadai.

Kriteria penerimaan teknis:

- Source diagram terverifikasi:
- `workspace/03\_Design-System/wireframes/sba-uc01-login-workspace-wireframe.mmd`
- `workspace/02\_Architecture/diagrams/sba-sequence-uc01-error-recovery.mmd`
- Render pipeline menghasilkan artefak tanpa error via `pnpm diagrams:build`.

8. Panduan Deployment Staging

Ringkas dari `docs/DEPLOYMENT\_STAGING.md`:

- Persiapan: lint, test, build
- Migrasi DB dan start services
- Rollback: stop, revert code/db, rebuild restart
- Verifikasi: health, logs, metrics, smoke test

Panduan implementasi teknis (deployment + verifikasi) tersedia pada `docs/SBA-Implementation-Guide.md`.

9. Referensi dan Lampiran

9.1 Referensi Dokumen

- `docs/README.md`, `docs/DEPLOYMENT\_STAGING.md`
- `docs/Use-Case Specification apps-app.md`
- Observability: `docs/TESTING-OBSERVABILITY.md`

Dokumen deliverable pendamping:

- Matriks fitur: ``docs/SBA-Feature-Matrix.md``
- Panduan implementasi: ``docs/SBA-Implementation-Guide.md``

9.2 Lampiran Diagram

- Mermaid blocks dan file ``*.mmd`` terkait (render via ``pnpm diagrams:build``)

9.3 Matriks Fitur

Berikut ringkasan mapping requirement vs implementasi (sumber: ``docs/SBA-Feature-Matrix.md``):

Requirement	Implementasi	Status	Dependensi
Auth (Supabase)	<code>`apps/app`</code> login, session/JWT	Implemented	Supabase URL/Key
RBAC	<code>`withRBAC`</code> , <code>`ensureTenantHeader`</code>	Implemented	Tenant context
Agents	<code>`features/agentic`</code> , AG-UI	Implemented	Orchestrator, <code>`@sba/rube`</code>
Runs	<code>`features/run-controls`</code> , <code>`/api/runs/*`</code>	Implemented	ToolRegistry
Analytics	<code>`features/analytics`</code> , metrics endpoints	Implemented	Observability/OTel
Monitoring	<code>`/monitoring`</code> , <code>`/metrics/*`</code>	Implemented	Prometheus
Onboarding	<code>`features/onboarding`</code>	Implemented	UX shell
Governance/Audit	audit store, RBAC guard	Implemented	Security headers